

DFT Properties and Applications

Signals and Linear Systems

Ashrafur Rahman
Lecturer, CSE, BUET

ashrafur@cse.buet.ac.bd

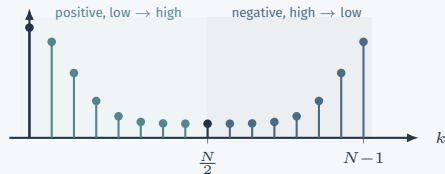
DFT / IDFT Pair ($W_N = e^{-j2\pi/N}$)

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$$

- **Three bridges:** $c_k = \frac{1}{T_0} X[k]$, $X[k] = N a_k$, and $X[k] = X(e^{j\omega})|_{\omega=\frac{2\pi}{N}k}$.
- Bin k lives at $f = k \frac{f_s}{N}$; spacing $\Delta f = \frac{f_s}{N}$.

$|X[k]|$



How to read N bins

Bins $0 \rightarrow \frac{N}{2}$: **positive** frequencies. Bins $\frac{N}{2} \rightarrow N$: folded **negative** frequencies. A real lowpass signal peaks at $k = 0$ and near $k = N$.

- **Part I — The rules of the game:** how the DFT behaves when a signal is shifted, flipped, modulated, or real — on N points everything wraps.

Linearity, circular shifts, modulation, conjugate symmetry, Parseval.

- **Part II — The payoff:** multiplying DFTs convolves signals — circularly, around the same clock. We will see the wrap, picture it, and then defeat it: zero-padding turns circular convolution into the linear convolution every filter needs.

One lecture, one theme

Last class we built the DFT. Today we learn its **grammar** — and cash it in: filtering signals by multiplying spectra.

The Rules of the Game

Everything Lives on a Clock

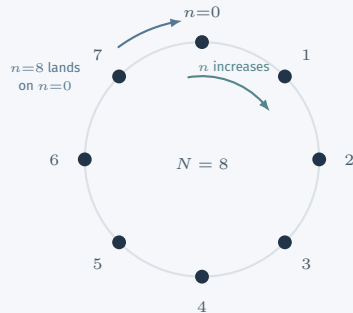
- The DFT only ever sees N numbers. Synthesis rebuilds them **N -periodically**: evaluating the IDFT at $n + N$ gives back $x[n]$. (Recall why: sampling in frequency, N bins, forced periodicity in time.)
- So indices live on a **clock**: define $((n))_N \triangleq n \bmod N$ — e.g. $((9))_8 = 1$ and $((-1))_8 = 7$.
- Tiling the N samples forever — one clock lap per copy — gives the **periodic extension**:

$$\tilde{x}[n] \triangleq x[((n))_N], \quad \tilde{x}[n + N] = \tilde{x}[n].$$

It matches $x[n]$ on $0, \dots, N - 1$ — the copy of x the DFT really works with.

The theme of Part I

Nothing falls off the end of N points — it **wraps**: every shift becomes a rotation of the clock.



Same clock for the bins k — we proved $X[k + N] = X[k]$ last time.

- Every property today comes straight from the analysis sum — nothing else is needed:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}.$$

- **Linearity.** The sum is linear in the samples, so $a x[n] + b y[n] \longleftrightarrow a X[k] + b Y[k]$.¹
- **Periodicity.** $W_N^{(k+N)n} = W_N^{kn}$, so $X[k + N] = X[k]$ — proved last lecture. The bins live on the same clock as the samples.
- Everything else in Part I — shifts, reversal, symmetry, energy — falls to a single move:

The one proof move of the day

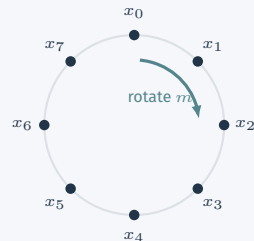
Substitute into the DFT sum, re-index, and let the clock absorb whatever spills past the edge: both $\tilde{x}[n]$ and W_N^{kn} are N -periodic, so **any window of N consecutive terms gives the same sum.**

¹Both sequences must have the same length N — pad the shorter one with zeros first. Remember this; it returns in Part II.

- Shift the periodic extension by m , take the DFT, substitute $r = n - m$:

$$\sum_{n=0}^{N-1} \tilde{x}[n - m] W_N^{kn} = W_N^{km} \sum_{r=-m}^{N-1-m} \tilde{x}[r] W_N^{kr} = W_N^{km} X[k].$$

- The window slid to $r = -m, \dots, N - 1 - m$ — but $\tilde{x}[r] W_N^{kr}$ is N -periodic, so any N consecutive terms give the same total: it is still $X[k]$. The wrap is invisible.



a shift is a **rotation**: every sample survives

Circular time shift

$$x[((n - m))_N] \longleftrightarrow W_N^{km} X[k] = e^{-j \frac{2\pi}{N} km} X[k]$$

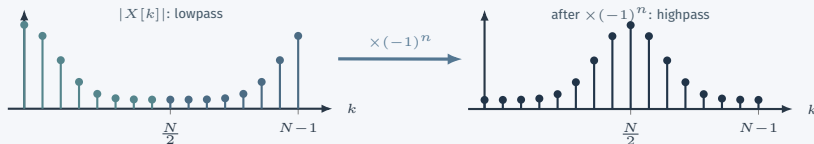
Phase only

$|W_N^{km}| = 1$: shifting changes **no magnitude**, only phase. Where a signal sits in time lives entirely in $\angle X[k]$.

- The mirror-image property — multiply by a complex tone in time, and the spectrum rotates:

$$W_N^{-k_0 n} x[n] = e^{j \frac{2\pi}{N} k_0 n} x[n] \longleftrightarrow X[((k - k_0))_N].$$

- Same one-line proof, run on the other side: absorb $W_N^{-k_0 n}$ into W_N^{kn} , and $k - k_0$ appears.
- Concrete special case, $k_0 = \frac{N}{2}$: $e^{j\pi n} = (-1)^n$, so **flipping every other sign** slides the spectrum by half the clock — low and high frequencies trade places:



- **Reversal** — playing the clock backwards (0 stays, $1 \leftrightarrow N-1$, $2 \leftrightarrow N-2$, ...) hands the minus sign to k . Substitute $r = -n$, then slide the window (the one move):

$$\sum_{n=0}^{N-1} \tilde{x}[-n] W_N^{kn} = \sum_{r=-(N-1)}^0 \tilde{x}[r] W_N^{(-k)r} = X[((-k))_N] \quad \Rightarrow \quad x[((-n))_N] \longleftrightarrow X[N-k].$$

- **Conjugation** — conjugating the sum flips the kernel to the $-k$ one, $(W_N^{kn})^* = W_N^{(-k)n}$:

$$\sum_{n=0}^{N-1} x^*[n] W_N^{kn} = \left(\sum_{n=0}^{N-1} x[n] W_N^{(-k)n} \right)^* = X^*[((-k))_N] \quad \Rightarrow \quad x^*[n] \longleftrightarrow X^*[N-k].$$

- Now let $x[n]$ be **real**: $x^*[n] = x[n]$, so its DFT must equal the conjugation result:

Real signal $\Rightarrow$ conjugate-symmetric spectrum

$$X[k] = X^*[((-k))_N] \quad \Longleftrightarrow \quad X[N-k] = X^*[k].$$

- Half the bins are forced by the other half. Next slide: what this does to $|X[k]|$ and $\angle X[k]$ — last lecture's bin-reading mantra, proved.

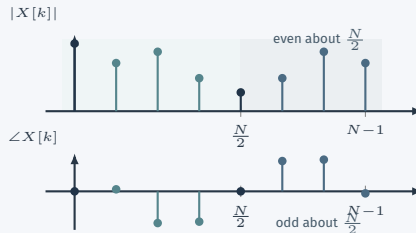
Conjugate Symmetry: Why Real Signals Fold

- We just proved $X[N - k] = X^*[k]$. In polar form, that is two symmetries about $k = \frac{N}{2}$:

$$|X[N - k]| = |X[k]| \quad \text{magnitude \textbf{even},}$$

$$\angle X[N - k] = -\angle X[k] \quad \text{phase \textbf{odd}.}$$

- Check on last lecture's $x = \{1, 2, 3, 4\}$, $X = \{10, -2+2j, -2, -2-2j\}$: indeed $X[3] = -2 - 2j = X^*[1]$. ✓
- Only bins $k = 0, \dots, \frac{N}{2}$ are free — the upper half is their mirror. (Corollary: real and even $\tilde{x} \Rightarrow X[k]$ real.)
- This is why the fold picture works:** bin $N - k$ is bin k 's negative-frequency partner.



the 8-point DFT of the real sequence $\{3, 5, 2, 0, 1, 2, 4, 1\}$

Parseval's relation

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$

- Proof in one substitution (sums run 0 to $N-1$): swap $x^*[n]$ for the conjugated IDFT, exchange the sums:

$$\sum_n x[n] x^*[n] = \sum_n x[n] \cdot \frac{1}{N} \sum_k X^*[k] W_N^{kn} = \frac{1}{N} \sum_k X^*[k] \underbrace{\sum_n x[n] W_N^{kn}}_{= X[k]} = \frac{1}{N} \sum_k |X[k]|^2.$$

- Check on $x = \{1, 2, 3, 4\}$: time side $1 + 4 + 9 + 16 = 30$; bin side $\frac{100+8+4+8}{4} = 30$. ✓
- So $\frac{|X[k]|^2}{N}$ is the **energy sitting in bin k** – the power spectrum.

Why it had to be true

The N tones W_N^{-kn} are **orthogonal**: the DFT is a change of basis in $\mathbb{C}^N$, and orthogonal moves re-bookkeep energy – they never create or destroy it.

The Property Table

Property	Time domain	DFT domain
linearity	$a x[n] + b y[n]$	$a X[k] + b Y[k]$
circular time shift	$x[((n - m))_N]$	$W_N^{km} X[k]$
modulation	$W_N^{-k_0 n} x[n]$	$X[((k - k_0))_N]$
time reversal	$x[((-n))_N]$	$X[((-k))_N]$
conjugation	$x^*[n]$	$X^*[((-k))_N]$
real signal	$x[n] \in \mathbb{R}$	$X[N - k] = X^*[k]$
duality	$X[n]$ (bins replayed as a signal)	$N x[((-k))_N]$
circular convolution	$x \circledast h$	$X[k] H[k] \rightarrow \text{Part II}$
multiplication	$x[n] h[n]$	$\frac{1}{N} X \circledast H$
Parseval	$\sum_n x[n] ^2$	$= \frac{1}{N} \sum_k X[k] ^2$

One table to keep

Every row follows from the DFT sum by the same move: re-index, and the clock absorbs the wrap. The two convolution rows are the rest of today.

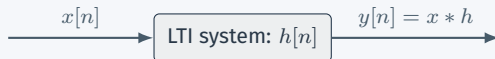
Filtering with the DFT

Why Convolution Matters

- From the first half of the course: an LTI system is completely described by its impulse response $h[n]$, and its output is the **linear convolution**

$$y[n] = (x * h)[n] = \sum_m x[m] h[n - m].$$

- Every filter you will ever run — smoothing, echo, equalizer — is this sum.
- Lengths add: x of length L , h of length $M \Rightarrow y$ of length $L + M - 1$. Direct cost: about $L \cdot M$ multiplications.



The dream

In every Fourier world so far, convolution in time became **multiplication** of spectra. If that holds for the DFT, filtering becomes: transform, multiply N numbers, transform back. Does it hold? Almost.

What Multiplying DFTs Actually Does

- Take two length- N sequences, multiply their DFTs bin by bin, and invert. What signal is that?

$$\begin{aligned} y[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] H[k] W_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] W_N^{km} \right) H[k] W_N^{-kn} \\ &= \sum_{m=0}^{N-1} x[m] \left(\frac{1}{N} \sum_{k=0}^{N-1} H[k] W_N^{-k(n-m)} \right) \quad \text{— the IDFT of } H, \text{ evaluated at } n - m. \end{aligned}$$

- But $n - m$ can be negative — and the IDFT evaluated off the grid returns the **periodic extension**:
 $W_N^{-k(n-m+N)} = W_N^{-k(n-m)}$, so the inner sum is $\tilde{h}[n - m] = h[((n - m))_N]$.

$$y[n] = \sum_{m=0}^{N-1} x[m] h[((n - m))_N].$$

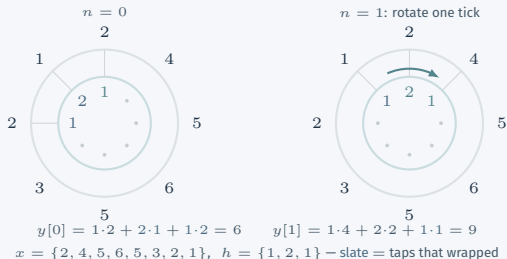
The wrap arrives uninvited

This is almost convolution — but the index into h wraps around the clock. Multiplying DFTs convolves **circularly**.

Circular convolution theorem (both sequences length N)

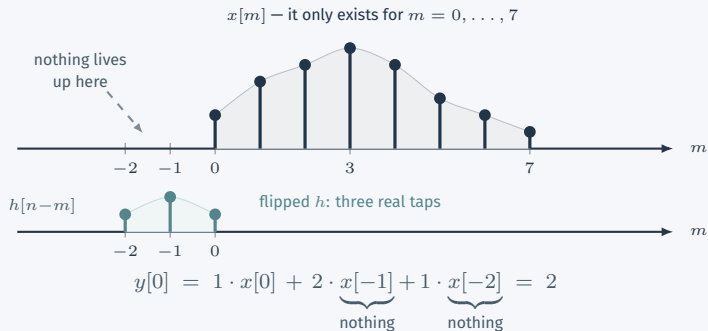
$$(x \circledast h)[n] \triangleq \sum_{m=0}^{N-1} x[m] h[(n-m)_N] \iff X[k] H[k]$$

- Flip, slide, multiply, add — but **on the clock**: x rides the outer wheel, h sits backwards on the inner. One output = rotate one tick, multiply across the spokes, add.



Step 1 — Linear: the Off-Edge Taps Find Nothing

- **Linear** convolution at $n = 0$: $y[0] = \sum_m x[m] h[0 - m]$. Flipping $h = \{1, 2, 1\}$ puts taps at $m = 0, -1, -2$:

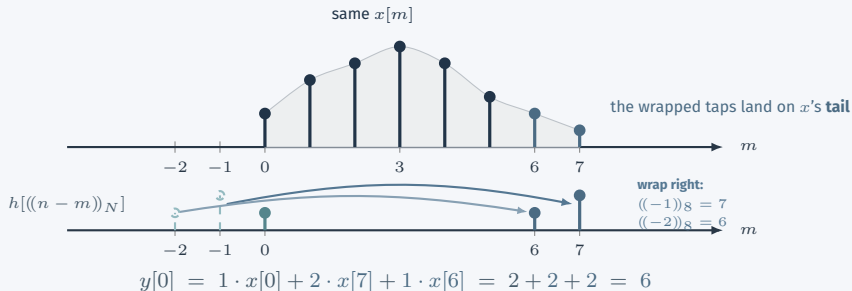


How linear convolution treats the edge

Taps hanging past the edge multiply **nothing** — their terms are silently dropped. (Hence the $L + M - 1$ -long output.)

Step 2 – Circular: the Taps Wrap to the Right

- The **circular** sum, $y[0] = \sum_m x[m] h[((0 - m))_N]$, reads indices on the clock: $((-1))_8 = 7$ and $((-2))_8 = 6$. The off-edge taps are not dropped — they are **moved**:

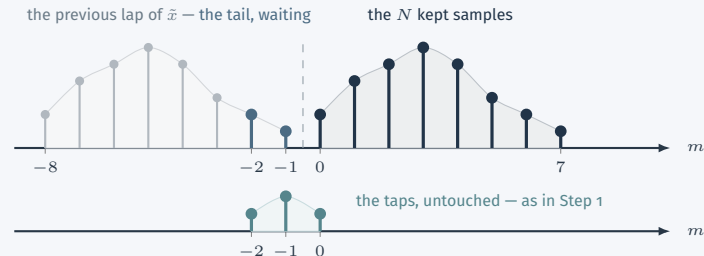


How the mod treats the edge

Nothing is dropped: every off-edge tap is **folded back** into $0, \dots, N - 1$ and multiplies the other end of x . That fold is the entire difference between circular and linear.

Step 3 – Equivalently: Extend x to the Left

- Same products, read the other way: **leave the taps where they fell**. The wrapped values $x[7], x[6]$ are exactly what the periodic extension $\tilde{x}$ puts at $m = -1, -2$:

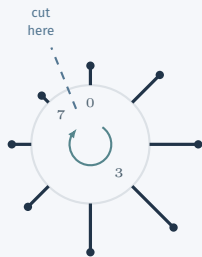


$$y[0] = 1 \cdot \tilde{x}[0] + 2 \cdot \tilde{x}[-1] + 1 \cdot \tilde{x}[-2] = 2 + 2 + 2 = 6 \quad \text{— identical}$$

Two readings of one sum

$\sum_m x[m] h[(n - m))_N] = \sum_j h[j] \tilde{x}[n - j]$: wrap the taps right, or extend x left. And periodically extending x is just... (next slide).

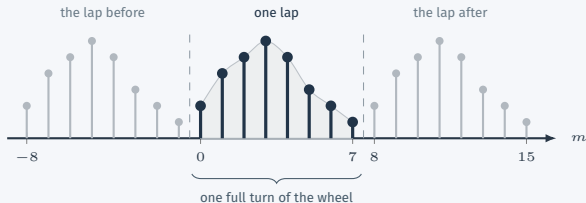
Unwinding the Clock



x , wrapped around the clock:

walking past $m = 7$ lands on $m = 0$

unroll
→



Same trip, two pictures

Going round and round the wheel is walking along the periodic extension $\tilde{x}$. So circular convolution has two equal readings: **rotate** the flipped h on the clock, or **slide** the flipped h along the unrolled $\tilde{x}$. Next, both compute our example.

Worked Example: Circular $\neq$ Linear

- Our running pair: $x = \{2, 4, 5, 6, 5, 3, 2, 1\}$,
 $h = \{1, 2, 1\}$; $L = 8, M = 3$.

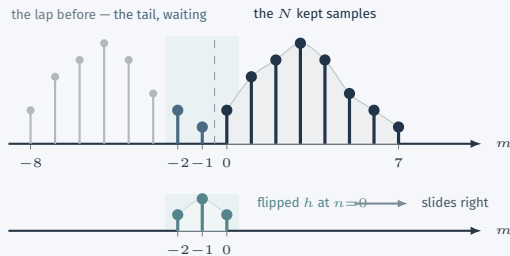
- **Linear** (length $8 + 3 - 1 = 10$):

$\{2, 8, 15, 20, 22, 19, 13, 8, 4, 1\}$

- **Circular** (length 8):

$x \circledast h = \{6, 9, 15, 20, 22, 19, 13, 8\}$

- Six values agree; the first **two** are corrupted — one per off-edge tap: on the unrolled clock, the sliding h grabs the previous lap's tail samples 1 and 2.



$$y[0] = 1 \cdot 2 + 2 \cdot 1 + 1 \cdot 2 = 6 \quad (\text{should have been } 2)$$

the window caught the previous lap's tail — that is the wrap

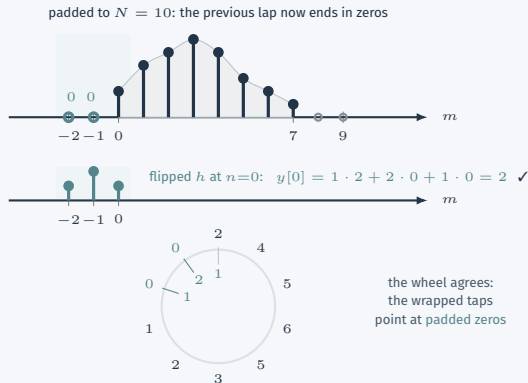
Zero-Padding Makes It Linear

- Circular convolution is the linear result folded mod N — **time aliasing**. If the linear result fits, nothing folds:

The zero-padding rule

Pad both sequences to length $N \geq L + M - 1$ and circular convolution **equals** linear convolution.

- Padded to $N = 10$, the circular result is the linear list: $\{2, 8, 15, \dots, 4, 1\}$. ✓
- The zeros aren't data — they are **room for the tail**. (Last lecture padding bought finer DTFT samples; today it buys correctness.)



Fast filtering with the DFT (input length L , filter length M)

1. Choose $N \geq L + M - 1$; zero-pad both x and h to length N .
 2. Compute $X[k]$ and $H[k]$ (N -point DFTs).
 3. Multiply bin by bin: $Y[k] = X[k] H[k]$.
 4. Invert: $y[n] = \text{IDFT}\{Y\}$; the first $L + M - 1$ samples are the linear convolution.
- Honest accounting, today: each DFT costs $\sim N^2$ multiplications, so this loses to the direct $L \cdot M$ sum. The transform is the bottleneck.
 - In `scipy`, `signal.fftconvolve` is this recipe. (For streaming audio, block variants like overlap-add run it chunk by chunk — advanced FFT lecture.)

Why this row of the table changed engineering

Next class the FFT drops each transform to $\sim N \log_2 N$ — and this recipe becomes **the fastest known way to convolve**.

The Big Picture

Properties: everything wraps

- $x[((n - m))_N] \leftrightarrow W_N^{km} X[k]$ (phase only)
- $W_N^{-k_0 n} x[n] \leftrightarrow X[((k - k_0))_N]$
- real $x[n]$: $X[N - k] = X^*[k]$ (the fold)
- $\sum_n |x[n]|^2 = \frac{1}{N} \sum_k |X[k]|^2$

The two theorems that matter

- $x \circledast h \leftrightarrow X[k]H[k]$, $x[n]h[n] \leftrightarrow \frac{1}{N} X \circledast H$
- circular = linear iff $N \geq L + M - 1$ (zero-pad!)

The pictures to remember

- the **clock**: indices mod N ; shifts are rotations
- **two wheels**: rotate, multiply, add = $x \circledast h$
- **sliding on $\tilde{x}$** : the neighbouring copy leaking into the window is the wrap; padding replaces it with zeros

Today in one sentence

On N points everything wraps — pad for the wrap, and the DFT turns filtering into multiplication.

Read

- Oppenheim, Willsky & Nawab, Problem 5.53 — the DFT, built as a worked exercise. (§5.3–5.5 state today's properties in DTFT notation, if you want a second telling.)
- Tan & Jiang, §4.1 — another telling of the DFT and its bins, with worked numbers.

Try

- Compare `np.convolve(x, h)` with `ifft(fft(x) * fft(h))` — unpadded, then padded to $L + M - 1$.
- Verify Parseval and conjugate symmetry on any real signal you like.

Next class

The DFT costs N^2 multiplications — for a one-second audio clip, that is already billions. Next class: the divide-and-conquer trick Gauss scribbled in 1805 and the world rediscovered in 1965 — the **Fast Fourier Transform**.